

HarmonyOS NEXT shim (Network Kit VPN)

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Revision: 2 Last modified: 2026-07-04T12:00:00Z **Rev 2:** Added an AppGallery review/distribution compliance note to §12 — closing a gap identified in an independent enterprise-hardening pass over Volume 4.

Master technical specification — Volume 4 (Clients), document v04-client/shim-harmonyos.md of the HelixVPN set. It **deepens** the HarmonyOS NEXT row of pass-1 doc 03 §5.5 (the Network Kit VPN *ability* shim) to nano-detail, implementation-ready granularity. It is SPEC-ONLY: it describes the exact shim to build — process topology, ArkTS ↔ NAPI ↔ Rust signatures, the TunnelPlatform contract impl, state machines, lifecycle, memory budgets, error handling, edge cases, and per-test-type coverage — but does not build it.

Boundary with sibling docs. This document **owns** the HarmonyOS NEXT tunnel shim only. It **consumes** (and must stay byte-consistent with): the Transport trait + helix-wg data plane (v02-data-plane/transport-trait.md), the orchestrator + frozen TunnelStatus broadcast enum (v02-data-plane/orchestrator-and-state.md §4.1), and the FFI surface + TunnelPlatform contract owned by doc 03 §3-§4. The Rust *internals* of helix-core are doc 01/02's; this doc owns only the HarmonyOS-specific binding + ability.

Evidence base. Citations inline by id: [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_UI] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [03_CLIENT] = pass-1 doc final/03-client-core-and-ui.md; [01-DP]/[01-ORCH] = final/v02-data-plane/*; [research-flutter_ffi] = the cited FFI/port research; [research-ios_android] = the sibling NE/VpnService shim research. Unproven platform facts are

marked **UNVERIFIED** per §11.4.6 (no-guessing); they are Phase-3 verification gates, never assumed parity with Android/iOS.

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0. Position, status, and the Phase-3 risk

HarmonyOS NEXT dropped Android/ART compatibility entirely — native ArkTS / ArkUI / DevEco only, so an Android APK or KMP artifact will not run on it [04_ARCH §5.2/§5.6]. The single-codebase path is the **OpenHarmony SIG Flutter fork** (gitee.com/openharmony-sig/flutter_flutter, ohos device channel → builds a .hap), engine recompiled for OHOS, tracking a **lagging Flutter baseline** (commonly cited Flutter OHOS ~3.22.x atop HarmonyOS SDK 5.0.0(12) / OpenHarmony API 10-12 — **UNVERIFIED**, pin at Phase-3 entry) [research-flutter_ffi §5].

The payoff/risk split [04_UI §6.1, 03_CLIENT §5.5]: *the UI ports for free, the shim does not*. Every Dart pixel (ConnectButton, ExitPicker, settings, account, policy) compiles unchanged; the irreducible per-platform work is this shim — an ArkTS VpnExtensionAbility driving the Network Kit VPN API, a NAPI bridge to the Rust .so, and a status relay. This is flagged the **biggest single platform risk in the program** (doc 03 §13 Phase 3, [04_UI §6.1]) for three reasons: (a) the SIG fork lags mainline and its plugin/embedder surface may lack a needed API (honest SKIP-with-reason per §11.4.3 is the fallback, never a faked PASS);

(b) HarmonyOS's VPN extension API + its memory/lifecycle limits are **UNVERIFIED** against our data-plane needs; (c) the toolchain (DevEco, signing) is a separate, China-hosted CI lane.

Property	Value	Source
Flutter fork	openharmy-sig/ flutter_flutter, ohos channel	[research-flutter_ffi §5]
Artifact	.hap (signed in DevEco Studio)	[research-flutter_ffi §5], [04_ARCH §5.2]
Rust core packaging	cdylib → libhelix_core.so (OHOS is Linux-kernel-based)	[research-flutter_ffi §3/§5]
Native bridge	N-API (Node-API) C++ module ↔ stable C ABI of the .so	[03_CLIENT §5.5], [research-flutter_ffi §5]
VPN mechanism	Network Kit VpnExtensionAbility + VpnConnection (ArkTS)	[research-flutter_ffi §5] (API names UNVERIFIED)
Phase	Phase 3 (second-tier target)	doc 03 §13

1. Process topology — split-process is the architecture

The load-bearing fact that shapes every signature below: on HarmonyOS NEXT the VPN runs in a **VpnExtensionAbility** — a separate ability, **almost certainly a separate OS process** from the UIAbility that hosts the Flutter engine (**UNVERIFIED** whether OHOS may co-locate them; design for separate-process and the co-located case degrades gracefully). This mirrors iOS's NEPacketTunnelProvider (separate extension process) and is **unlike** Android's in-process VpnService [research-ios_android]. Consequence: the data-plane core (the Rust .so + orchestrator, the status authority per O-I10) lives in the **ability process**, so its frb StreamSink<TunnelStatus> **cannot** reach the Dart isolate in the UI process directly. Status must be **relayed** across the process boundary (§7), then re-emitted on the Dart status stream so the UI contract (CI2: *the UI is a pure function of the status stream*) is preserved physically as well as logically.

[MERMAID RENDER FAILED — block #1]

The seam (identical in spirit to doc 03 §3.3): **lifecycle commands flow UI → shim → core** (startTunnel/stopTunnel cross the IPC into the ability), while **status flows core → relay → UI** (the core owns truth; the relay carries it back). setShields/setExit/exits are pure-

logic FFI calls **but** because the orchestrator lives in the ability process, on HarmonyOS they are *also* marshalled across the IPC as ability commands (§5.4), unlike Android where they hit the in-process core directly.

2. The contract this shim satisfies (TunnelPlatform)

The HarmonyOS shim implements **exactly** the Dart TunnelPlatform contract from doc 03 §4 — one MethodChannel("helixvpn/tunnel") + one EventChannel("helixvpn/tunnel/events"). No HarmonyOS type leaks above the channel; everything above is shared Dart.

```
// packages/helix_core_ffi/lib/tunnel_platform_ohos.dart
// The OHOS realization of the abstract TunnelPlatform (doc 03 §4).
class OhosTunnelPlatform implements TunnelPlatform {
  static const _m = MethodChannel('helixvpn/tunnel');
  static const _e = EventChannel('helixvpn/tunnel/events');

  @override
  Future<void> startTunnel(TunnelConfig cfg) async {
    // cfg fields are doc 03 §4: overlayIp, routes, dnsServers,
    // splitExcludeApps,
    // mtu, sessionOrMapToken. Marshalled to a map the ArkTS host
    // decodes (§6).
    await _m.invokeMethod('startTunnel', cfg.toMap());
  }

  @override
  Future<void> stopTunnel() => _m.invokeMethod('stopTunnel');

  @override
  Stream<PlatformTunnelEvent> events() => _e
    .receiveBroadcastStream()
    .map(PlatformTunnelEvent.fromMap); // up|down|
    permissionDenied|revoked|error
}
```

PlatformTunnelEventKind is the doc 03 §4 closed set — up, down, permissionDenied, revoked, error. The EventChannel carries **only** lifecycle (not TunnelStatus, not packet data); TunnelStatus arrives on the separate frb status stream fed by the §7 relay.

3. The stable C ABI (helix-core exported surface)

All native shims (iOS cbindgen, Android JNI thunk, HarmonyOS NAPI) bind to one **stable C ABI** exported from the helix-ffi crate via cbindgen (crate-type = ["staticlib", "cdylib"]; HarmonyOS consumes the cdylib libhelix_core.so). This is the C projection of the frb/UniFFI surface in doc 03 §3.1 and is consistent with the iOS helix_core_* symbols seen in [03_CLIENT §5.1]. The HarmonyOS shim uses the **fd-pump model** (like Android's ParcelFileDescriptor, unlike iOS's packetFlow callback) because VpnConnection.create returns a tun **file descriptor** — so the core pumps the fd itself in-process (in the ability process).

```
/* helix_core.h – generated by cbindgen from crates/helix-ffi.
   Stable C ABI.
   * Thread-safety: all functions are safe to call from the
     ability's main ArkTS
   * worker thread; the core spawns its own tokio runtime threads
     internally.
   * Every int return: 0 = OK, <0 = HelixErr (see §3.1). */

#include <stdint.h>
#include <stddef.h>

/* ---- handle lifecycle ---- */
/* mode: 0 = Client, 1 = Connector (doc 03 CoreMode).
   session_or_map = NUL-term
   * session token (Phase 1+) or map file path (Phase 0). Returns a
     handle id >0,
   * or <0 HelixErr. Idempotent guard: a second start without stop
     returns
   * HELIX_ERR_ALREADY_STARTED. */
int64_t helix_core_start(const char* session_or_map, int32_t
mode);

/* Graceful stop of handle h: cancels loops, FINs transport,
   reverts
   * kill-switch+DNS, emits Down{reason="stopped"} on the status
     callback, frees h.
   * Idempotent: stopping an unknown/stopped handle returns 0. */
int32_t helix_core_stop(int64_t h);

/* ---- TUN handoff (CI1: the core never opens the TUN; the
   ability does) ---- */
/* Hand the core an OS tun fd from VpnConnection.create. The core
   dup()'s it and
   * owns the dup; caller may close its copy. Starts the in-process
     pump. */
int32_t helix_core_attach_tun(int64_t h, int32_t tun_fd, uint32_t
mtu);
int32_t helix_core_detach_tun(int64_t h);
```

```

/* ---- status callback (the relay source, §7) ---- */
/* cb receives a UTF-8 JSON encoding of TunnelStatus (§9) on every
   transition.
   * ctx is opaque, passed through. Called from a core thread – the
     C++ bridge MUST
   * forward via a NAPI threadsafe-function, never touch the ArkTS
     VM directly. */
typedef void (*helix_status_cb)(const char* status_json, void*
    ctx);
int32_t helix_core_set_status_cb(int64_t h, helix_status_cb cb,
    void* ctx);

/* ---- logic-only ops (doc 03 §3.1) – marshalled JSON in/out ----
   */
/* exits(): caller frees the returned malloc'd JSON via
   helix_string_free. */
char* helix_core_exits(int64_t
    h); /* JSON Vec<ExitOption> */
int32_t helix_core_set_exit(int64_t h, const char* exit_json); /
    * {id, multiHopChain?} */
int32_t helix_core_set_shields(int64_t h, const char*
    shields_json); /* Shields (doc 03 §3.1) */
char* helix_core_advertise(int64_t h, const char* cidrs_json); /
    * connector mode; JSON AdvertiseResult */
void helix_string_free(char* s);

/* ---- diagnostics ---- */
const char* helix_core_version(void); /* static, do not free */
const char* helix_last_error(void); /* thread-local last
    HelixErr detail; static lifetime per call */

```

3.1 HelixErr closed set (negative C returns)

Mirrors the orchestrator's Down.reason vocabulary [01-ORCH §4.4] plus shim-startup errors, so a numeric code maps to a stable string (§11.4.6).

C const	int	Meaning	Relayed Down.reason
HELIX_ERR_ALREADY_STARTED	-1	start on a live handle	(no Down)
HELIX_ERR_BAD_CONFIG	-2	unparseable session/map/json	host-fatal
HELIX_ERR_TUN_ATTACH	-3	attach_tun dup/ ioctl failed	host-fatal
HELIX_ERR_AUTH	-4	WG handshake rejected	auth-failed
HELIX_ERR_NO_ROUTE	-5	RouteMap has no peer	no-route
HELIX_ERR_LADDER_EXHAUSTED	-6	every transport rung failed	ladder-exhausted

C const	int	Meaning	Relayed Down . reason
HELIX_ERR_PINNED_FAILED	-7	pinned transport failed	pinned-transport-failed
HELIX_ERR_INTERNAL	-99	unexpected; detail in helix_last_error()	host-fatal

4. The NAPI bridge layer (C++ ↔ Rust .so)

DevEco's native module template produces a CMake .so that registers N-API functions. libhelix_napi.so is that module — a thin, **stateless-per-call** C++ translator between ArkTS values and the §3 C ABI. It owns one global: the NAPI **threadsafe-function (TSFN)** used to ferry status JSON from a core thread into the ArkTS event loop (§7). No business logic lives here.

```
// entry/src/main/cpp/helix_napi.cpp - HarmonyOS NAPI module for
// the ability.
#include "napi/native_api.h"
#include "helix_core.h"
#include <cstring>

static napi_threadsafe_function g_status_tsfm = nullptr; // set
// by registerStatus()

// ---- napi_value Start(env, [sessionOrMap: string, mode:
// number]): bigint(handle) ----
static napi_value Start(napi_env env, napi_callback_info info) {
    size_t argc = 2; napi_value args[2];
    napi_get_cb_info(env, info, &argc, args, nullptr, nullptr);
    char sess[2048]; size_t n = 0;
    napi_get_value_string_utf8(env, args[0], sess, sizeof(sess),
        &n);
    int32_t mode = 0; napi_get_value_int32(env, args[1], &mode);
    int64_t h = helix_core_start(sess, mode); // §3
        C ABI
    napi_value out; napi_create_bigint_int64(env, h, &out);
    return out; // <0
        => ArkTS checks HelixErr
}

// ---- AttachTun(env, [handle: bigint, fd: number, mtu: number]):
// number ----
static napi_value AttachTun(napi_env env, napi_callback_info
    info) {
```

```

    size_t argc = 3; napi_value a[3];
    napi_get_cb_info(env, info, &argc, a, nullptr, nullptr);
    int64_t h; bool lossless; napi_get_value_bigint_int64(env,
        a[0], &h, &lossless);
    int32_t fd, mtu; napi_get_value_int32(env, a[1], &fd);
        napi_get_value_int32(env, a[2], &mtu);
    int32_t rc = helix_core_attach_tun(h, fd, (uint32_t)mtu);
    napi_value out; napi_create_int32(env, rc, &out); return out;
}

// ---- the C status callback runs on a CORE thread → bounce to
//       TSFN (never touch VM here)
static void status_trampoline(const char* json, void* /*ctx*/) {
    if (!g_status_tsfn) return;
    char* dup = strdup(json); //
        freed in js_status_cb
    napi_call_threadsafe_function(g_status_tsfn, dup,
        napi_tsfn_nonblocking);
}
// js_status_cb (omitted) runs on the ArkTS loop: builds a JS
// string, invokes the
// ArkTS callback, then free(dup). Registered via
// RegisterStatus(handle, jsCb).

EXTERN_C_START
static napi_value Init(napi_env env, napi_value exports) {
    napi_property_descriptor desc[] = {
        {"start", nullptr, Start, nullptr, nullptr, nullptr,
            napi_default, nullptr},
        {"stop", nullptr, Stop, nullptr, nullptr, nullptr,
            napi_default, nullptr},
        {"attachTun", nullptr, AttachTun, nullptr, nullptr, nullptr,
            napi_default, nullptr},
        {"detachTun", nullptr, DetachTun, nullptr, nullptr, nullptr,
            napi_default, nullptr},
        {"registerStatus", nullptr, RegisterStatus, nullptr, nullptr,
            nullptr, napi_default, nullptr},
        {"setShields", nullptr, SetShields, nullptr, nullptr,
            nullptr, napi_default, nullptr},
        {"setExit", nullptr, SetExit, nullptr, nullptr, nullptr,
            napi_default, nullptr},
        {"exits", nullptr, Exits, nullptr, nullptr, nullptr,
            napi_default, nullptr},
    };
    napi_define_properties(env, exports, sizeof(desc)/
        sizeof(desc[0]), desc);
    return exports;
}
EXTERN_C_END
static napi_module helixMod = { .nm_version = 1, .nm_flags =
    0, .nm_filename = nullptr,
    .nm_register_func = Init, .nm_modname = "helix_napi", .nm_priv
    = nullptr, {0} };
extern "C" __attribute__((constructor)) void RegisterHelix(void)
{ napi_module_register(&helixMod); }

```


N-API rationale [03_CLIENT §5.5, research-flutter_ffi §2/§5]: rather than uniffi-dart (community, "should not be trusted") or a HarmonyOS-native UniFFI path, the shim binds the **stable cbindgen C ABI** through a hand-written ~200-LOC N-API module. This keeps **one** Rust surface (frb for Dart, cbindgen C for all native shims) and avoids a fourth binding generator. The exact ArkTS↔C++ N-API import path (import helix from 'libhelix_napi.so') and the oh-package.json5 native-module declaration are DevEco conventions — **UNVERIFIED** version specifics, pin at Phase-3 entry.

5. The ArkTS VpnExtensionAbility (the shim core)

The ability is the HarmonyOS analogue of iOS PacketTunnelProvider / Android VpnService. It (a) requests VPN permission + creates the VpnConnection, (b) hands the tun fd to the core via NAPI, (c) registers the status relay, (d) reports lifecycle on the EventChannel. **API names below are the OpenHarmony Network Kit VPN surface (@ohos.net.vpnExtension) as cited by [research-flutter_ffi §5] — exact method signatures are UNVERIFIED and MUST be pinned against DevEco SDK docs at Phase-3 entry (§11.4.99 latest-source).**

```
// entry/src/main/ets/serviceability/HelixVpnAbility.ets
import VpnExtensionAbility from
    '@ohos.app.ability.VpnExtensionAbility';    // UNVERIFIED
    path
import vpnExt from
    '@ohos.net.vpnExtension';                    //
    UNVERIFIED path
import helix from
    'libhelix_napi.so';                          //
    §4 N-API module
import type Want from '@ohos.app.ability.Want';

interface HelixTunnelCfg {                      // decoded from the
    MethodChannel map (doc 03 §4)
    overlayIp: string; routes: string[]; dnsServers: string[];
    splitExcludeApps: string[]; mtu: number; sessionOrMapToken:
        string;
}

export default class HelixVpnAbility extends VpnExtensionAbility {
    private conn: vpnExt.VpnConnection | null = null;
    private handle: bigint = 0n;                // helix_core handle (§3);
    0n = stopped
```

```

// Phase A – permission + tun creation + core start. Invoked by
// the cross-ability
// start command (§7). cfg arrives marshalled from Dart (§6).
async onStart(want: Want, cfg: HelixTunnelCfg): Promise<void> {
  this.conn =
    vpnExt.createVpnConnection(this.context); //
    UNVERIFIED
  const cfgBuilder: vpnExt.VpnConfig = {
    addresses: [{ address: { address: cfg.overlayIp, family:
      1 }, prefixLength: 32 }],
    routes: cfg.routes.map(r => parseRoute(r)), //
      AllowedIPs → OS routes
    dnsAddresses: cfg.dnsServers, // tunnel
      DNS (anti-leak)
    mtu: cfg.mtu, // 1280
      MASQUE / 1420 plain WG
    // splitExcludeApps → trustedApplications/
    // blockedApplications (UNVERIFIED field name)
    blockedApplications: cfg.splitExcludeApps,
  };
  let tunFd: number;
  try {
    tunFd = await this.conn.create(cfgBuilder); // returns
      tun fd – UNVERIFIED return shape
  } catch (e) {
    this.emitLifecycle(isPermDenied(e) ? 'permissionDenied' :
      'error', `${e}`);
    return;
  }
  this.handle = helix.start(cfg.sessionOrMapToken, 0 /
    *Client*/); // §4
  if (this.handle <= 0n) { this.emitLifecycle('error',
    `core_start ${this.handle}`); return; }
  helix.registerStatus(this.handle, (json: string) =>
    this.relayStatus(json)); // §7
  const rc = helix.attachTun(this.handle, tunFd,
    cfg.mtu); // §4 fd-pump
  if (rc !== 0) { this.emitLifecycle('error', `attach_tun ${rc}`
    `); await this.onStop(); return; }
  this.emitLifecycle('up', 'tun-
    attached'); // EventChannel up
}

async onStop(): Promise<void> {
  if (this.handle > 0n) { helix.detachTun(this.handle);
    helix.stop(this.handle); this.handle = 0n; }
  try { await this.conn?.destroy(); } catch (_)
    { /* best-effort */ } // UNVERIFIED
  this.conn = null;
  this.emitLifecycle('down', 'stopped');
}

// OS-initiated revoke (another VPN took over / user revoked
// permission):

```

```

onConnectionRevoked(): void { this.emitLifecycle('revoked', 'os-
revoked'); void this.onStop(); }

// logic-only commands marshalled in from the UI process (§5.4):
setShields(json: string): number { return this.handle > 0n ?
    helix.setShields(this.handle, json) : -2; }
setExit(json: string): number { return this.handle > 0n ?
    helix.setExit(this.handle, json) : -2; }
exits(): string { return this.handle > 0n ?
    helix.exits(this.handle) : '[]'; }

private relayStatus(statusJson: string): void { /* §7 – push to
    UI process */ }
private emitLifecycle(kind: string, detail: string): void { /*
    §7 – push to EventChannel */ }
}

```

5.4 Why logic ops cross the IPC on HarmonyOS

On Android the orchestrator is in-process, so `core.setShields(...)` is a direct frb call (doc 03 §3.3). On HarmonyOS the orchestrator lives in the **ability** process, so `setShields/setExit/exits/advertise` MUST be marshalled across the cross-ability IPC (§7) into `HelixVpnAbility`, which forwards to NAPI. The Dart `HelixCore` wrapper (doc 03 §3.2) is unchanged; only its OHOS *implementation* routes through the platform channel instead of frb-in-process. This is the one place the HarmonyOS FFI wiring deviates from the doc 03 §3.3 "logic goes straight UI→FFI" rule — documented here so the deviation is explicit, not silent (§11.4.6).

6. The UIAbility side — MethodChannel/EventChannel plumbing

The Flutter engine is hosted in a `UIAbility`. The SIG fork's `FlutterAbility/FlutterEngine` exposes the standard `MethodChannel/EventChannel` host APIs in ArkTS (the OHOS equivalent of Android's `MethodChannel` registrar — **UNVERIFIED** exact class names, pin at Phase-3). The `UIAbility` glue registers the two channels and bridges them to the ability via the §7 IPC.

```

// entry/src/main/ets/entryability/EntryAbility.ets (UIAbility,
    hosts Flutter)
import { MethodChannel, EventChannel } from '@ohos/
    flutter_ohos'; // UNVERIFIED package
import { HelixVpnClient } from '../relay/
    HelixVpnClient'; // §7 IPC client

export function registerHelixChannels(messenger:
    BinaryMessenger): void {

```

```

const client = new
    HelixVpnClient(); //
    talks to the ability
const m = new MethodChannel(messenger, 'helixvpn/tunnel');
m.setMethodCallHandler({
    onMethodCall: async (call, result) => {
        switch (call.method) {
            case 'startTunnel': await client.start(call.args as
                Record<string, Object>); result.success(null); break;
            case 'stopTunnel': await client.stop();
                result.success(null); break;
            case 'setShields': result.success(await
                client.setShields(call.args as string)); break;
            case 'setExit': result.success(await
                client.setExit(call.args as string)); break;
            case 'exits': result.success(await client.exits());
                break;
            default: result.notImplemented();
        }
    },
});
const e = new EventChannel(messenger, 'helixvpn/tunnel/events');
e.setStreamHandler({
    onListen: (_args, sink) =>
        client.bindLifecycleSink(sink), // up|down|
        permissionDenied|revoked|error
    onCancel: (_args) => client.unbindLifecycleSink(),
});
}

```

The TunnelConfig.toMap() (Dart) $\rightleftharpoons$ HelixTunnelCfg (ArkTS) marshalling keys are the doc 03 §4 field names verbatim (overlayIp, routes, dnsServers, splitExcludeApps, mtu, sessionOrMapToken) — a contract test (§13) asserts the round-trip is lossless.

7. The status relay (preserving CI2 across the process boundary)

The core's TunnelStatus originates in the ability process; the UI must see it on the frb status stream (CI2). The relay is the bridge. Two physical transports are candidates — pick one at Phase-3 entry; **the choice is UNVERIFIED** and is open decision **D-OHOS-1** (§14):

Candidate IPC	Pros	Cons
@ohos.rpc (IRemoteObject, ability connect)	typed, request/ response + callback	most boilerplate; lifecycle binding

Candidate IPC	Pros	Cons
@ohos.commonEventManager (publish/subscribe)	simplest fan-out for status events	best-effort delivery; ordering not guaranteed
@ohos.distributedKVStore / shared storage watch	survives ability restart	latency; not event-shaped

Recommended (UNVERIFIED): @ohos.rpc — the ability exposes an IRemoteObject; the UIAbility's HelixVpnClient connects to it, sends start/stop/setShields/... requests, and registers a **callback object** the ability invokes on every status transition and lifecycle event. RPC ordering + delivery guarantees are what the single-state-authority contract (O-I10) needs.

[MERMAID RENDER FAILED — block #2]

The frb status re-emitter (FFIr) is a thin Dart sink: the OHOS build feeds the relayed JSON into the same Stream<TunnelStatus> the rest of the app watches, so **no UI code is platform-aware** (CI2 holds). On single-process platforms the frb StreamSink feeds this stream directly; on HarmonyOS the relay feeds it — the *shape* is identical.

8. Lifecycle state machines

8.1 Ability lifecycle (shim-owned)

[MERMAID RENDER FAILED — block #3]

The ability state machine is **lifecycle only**; it never invents a TunnelStatus. Protection-state truth (Connecting/Handshaking/Connected/...) is the orchestrator's (O-I10), surfaced via the §7 relay, never synthesized by the ability. This is the seam that prevents "UI says connected while the tunnel is down" (doc 03 §3.3).

8.2 Kill-switch ordering across the process boundary

Per O-I11 (kill-switch fails closed), the firewall side-effects are the orchestrator's and execute **in the ability process before the status emit**: on any transition into Reconnecting/Down, the core closes the firewall, *then* relays the status. The relay carries already-applied truth, so a UI reacting to Connected can trust the tunnel is open and a UI reacting to Danger/Down knows the firewall is already closed. The ability process crashing leaves the OS

VpnConnection torn down by the OS (fail-closed by construction — no VpnConnection, no route); the UIAbility observes the RPC death and surfaces PlatformTunnelEvent.down (§11).

9. TunnelStatus consistency mapping

The frozen orchestrator broadcast enum is the 5-variant contract [01-ORCH §4.1]:

```
// crates/helix-core – the SINGLE state authority (0-I10). FROZEN.
pub enum TunnelStatus {
    Connecting,
    Handshaking,
    Connected { transport: String, rtt_ms: u32 },
    Reconnecting,
    Down { reason: String }, // reason vocab: §3.1 / [01-ORCH §4.4]
}
```

The **FFI-facing** enum (doc 03 §3.1) mirrors-and-extends it with three client-only additions — Disconnected (clean idle, distinct from Down), path on Connected ("direct"|"relay"), and Danger{kind} (leak / kill-switch tripped). The HarmonyOS relay carries the **FFI-facing** JSON, so the OHOS build's Stream<TunnelStatus> is byte-identical to every other platform's. The mapping (orchestrator → FFI projection) the relay applies:

Orchestrator OrchState/ TunnelStatus	Relayed FFI TunnelStatus JSON
Idle (internal)	{"t":"Disconnected"}
Connecting	{"t":"Connecting"}
Handshaking	{"t":"Handshaking"}
Connected{transport,rtt_ms}	{"t":"Connected","transport":...,"path":...,"rtt_ms":...}
Reconnecting{...}	{"t":"Reconnecting"}
ShuttingDown (internal)	{"t":"Down","reason":"stopped"}

Orchestrator OrchState/ TunnelStatus	Relayed FFI TunnelStatus JSON
Down{reason}	{"t": "Down", "reason": ...}
(leak / kill-switch tripped)	{"t": "Danger", "kind": "leak" "killswitch_tripped"}

Consistency gate (§13 unit/contract test): the ArkTS relay's JSON tag set and the Dart TunnelStatus frb mirror MUST match the orchestrator enum + the three documented extensions exactly. A drift (extra/missing tag, renamed field) FAILs the contract test — the same byte-for-byte assertion doc 03 §3.1 mandates for the frb mirror, here applied to the relay JSON codec.

10. Memory & resource budgets

The iOS Network Extension working-set ceiling (historically ~15 MB) is the program's hardest constraint and the reason the core is Rust (CI4) [04_ARCH §5.6, research-ios_android]. **HarmonyOS's VpnExtensionAbility process memory ceiling is UNVERIFIED** — it is a Phase-3 G-gate measurement, not an assumption. Design targets (mirroring the iOS discipline so the OHOS build inherits the lean profile for free):

Budget	Target	Verification (Phase 3)
Ability-process steady-state RSS	$\leq$ ~16 MB design target (actual ceiling UNVERIFIED)	1 GB sustained transfer on a real HarmonyOS device, sample ability-process RSS via DevEco profiler
Rust .so build profile	opt-level=z + thin-LTO + panic=abort + strip, aarch64 OHOS target	matches the iOS G3 recipe [research-ios_android]
MASQUE vs plain-UDP RSS	both pass separately (QUIC buffers cost more)	run each transport in isolation, $\geq 30\%$ headroom over the device ceiling across a 30-min soak
NAPI bridge overhead	~constant; no per-packet allocation (fd-pump, not callback)	TSFN used only for status ($\ll 1$ msg/s), never packets
.hap size		

Budget	Target	Verification (Phase 3)
	within the doc 03 §10 client size budget	DevEco artifact inspection

Fallback ladder if the ability ceiling is breached (each a real product decision, never hand-waved — mirrors the iOS §5.1 fallbacks): (1) shrink buffers / cap QUIC flow-control windows in the OHOS build; (2) move MASQUE off-device for HarmonyOS (plain WG + on-path obfuscation only — partial feature loss); (3) split so only the lean WG datapath is in the ability and QUIC negotiation lives in the UI process via the RPC channel. The chosen rung is **D-OHOS-2** (§14).

11. Error handling & edge cases

#	Edge case	Detection	Handling
E1	VPN permission denied	VpnConnection.create throws perm error	PlatformTunnelEvent.permissionDenied; no core start; UI prompts re-grant
E2	Another VPN active / OS revokes	onConnectionRevoked	emit revoked, teardown, kill-switch already closed (O-I11)
E3	attach_tun fails (bad fd)	helix_core_attach_tun returns HELIX_ERR_TUN_ATTACH	emit error, run onStop, surface Down{host-fatal}
E4	Ability process crash	UIAbility RPC death callback	PlatformTunnelEvent.down; OS auto-tears VpnConnection (fail-closed); UI offers reconnect
E5	UI process killed, tunnel alive	OS keeps the ability running	on UI relaunch, HelixVpnClient re-binds RPC + re-subscribes status; relay replays last TunnelStatus (latest-wins, not history)
E6	Status callback after stop	core stopped but a late helix_status_cb fires	TSFN guarded by handle generation; stale-handle status dropped (no UI flicker)
E7	WG auth rejected	core returns Down{auth-failed}	relay → Danger? No — Down{auth-failed}; UI shows re-enroll path; kill-switch stays closed
E8	DNS leak / kill-switch trip	core emits Danger{kind}	relay → red palette; firewall already closed before emit (§8.2)
E9	SIG-fork lacks a channel API	build/integration time	honest SKIP-with-reason per §11.4.3 + tracked migration item; never a faked PASS
E10			

#	Edge case	Detection	Handling
	MTU mismatch (1280 MASQUE vs 1420 plain)	negotiated mtu in TunnelConfig	passed to both VpnConfig.mtu and attach_tun(mtu); mismatch is a contract-test failure (§13)
E11	Split-tunnel app list unsupported	blockedApplications field absent in SIG SDK	SKIP-with-reason; degrade to full-tunnel + surface to user (no silent drop, §11.4.6)
E12	TSFN backpressure	status burst	napi_tsfm_nonblocking drops oldest on full queue; status is latest-wins so a dropped intermediate is harmless (final state always delivered)

All helix_last_error() strings are logged **without** the session token / map contents (§11.4.10 credentials never logged).

12. Build, signing, packaging

Step	Tool / artifact	Notes
Rust core	cargo build --target aarch64-unknown-linux-ohos (OHOS target via ohos-rs/NDK — UNVERIFIED target triple, pin Phase-3) → libhelix_core.so	cdylib; lean profile (§10)
C ABI header	cbindgen → helix_core.h	shared with iOS/Android shims (§3)
NAPI module	DevEco CMake native module → libhelix_napi.so (§4)	links libhelix_core.so
Flutter engine	openharmony-sig/flutter_flutter, ohos channel	pinned baseline (§0)
App assembly	flutter build hap (SIG fork) → DevEco project	embeds both .sos under the ability module
Signing	DevEco Studio (HarmonyOS app cert/profile)	separate CI lane; cert material per §11.4.10 (gitignored, never logged)
module.json5	declares VpnExtensionAbility + ohos.permission.VPN-class	pin Phase-3

Step	Tool / artifact	Notes
	permission (UNVERIFIED exact permission id)	

The OHOS target triple, the VPN permission identifier, and the oh-package.json5 native-module declaration are **UNVERIFIED** DevEco/SDK specifics — each is a Phase-3 latest-source verification (§11.4.99), not a memorized constant.

12.1 AppGallery review + distribution (engineering-relevant subset)

Huawei AppGallery applies its own review to apps declaring a VPN-class ability (VpnExtensionAbility) — an engineering-schedulable dependency of the Phase-3 plan, not a submission-day surprise: (a) the ohos.permission.VPN-class permission (exact identifier **UNVERIFIED**, §12) is a sensitive-permission category subject to functional-claim review (the app must genuinely establish a VPN, not merely declare the permission); (b) a privacy-disclosure + data-handling statement consistent with the no-durable-traffic-log architecture (v05-security/no-logging-as-code.md) is required, mirroring the Play/App-Store data-safety-form expectation on the other two mobile platforms; (c) HarmonyOS app signing/certification runs through Huawei's own AppGallery Connect + DevEco pipeline (§12), a separate credential + lead-time chain from the Apple/Google/Microsoft ones. Recommendation: budget AppGallery review lead time into the Phase-3 exit-gate schedule (09-phase3-reach-wbs.md) alongside the HarmonyOS shim implementation effort itself, since the two are independent critical-path items (implementation can finish while review is still pending).

13. Test points — §11.4.169 comprehensive test-type coverage

Per the LIVE constitution anchor §11.4.169 (mandatory comprehensive test-type coverage) composed with §11.4.27 (100 % test-type coverage) and §11.4.3 (topology-aware SKIP-with-reason where the SIG embedder genuinely lacks an API — never a faked PASS), every shim surface carries the applicable test types with **real captured evidence** (§11.4.5/§11.4.69) — no metadata-only PASS.

Test type	What it exercises here	Evidence / anti-bluff
Unit	NAPI value marshalling (string/bigint/int round-trips, §4);	golden-good/golden-bad fixtures

Test type	What it exercises here	Evidence / anti-bluff
Contract / consistency	TunnelConfig.toMap ⇌ HelixTunnelCfg; HelixErr→Down.reason map (§3.1); relay JSON codec ⇌ frb TunnelStatus mirror (§9). Riverpod fakes via StreamProvider.overrideWithValue [research-flutter_ffi §6]. the §9 byte-for-byte TunnelStatus assertion across orchestrator enum ⇌ relay JSON ⇌ Dart mirror; the doc 03 §4 TunnelPlatform channel signature.	(§11.4.107(10)); a renamed enum tag FAILs the §9 consistency gate. drift = FAIL (mirrors doc 03 §3.1 frb-mirror test).
Integration	ArkTS ability ⇌ NAPI ⇌ real libhelix_core.so against a real local gateway: start → attach_tun → Connected over plain-UDP and MASQUE; setShields/setExit/exits across the RPC (§5.4). No mocks beyond unit (§11.4.27).	captured status timeline + sink-side evidence (§11.4.69 network_connectivity).
E2E (full-automation)	real device: launch HAP → grant VPN permission → connect → real traffic flows through the overlay → toggle kill-switch → leak check → stop; fully self-driving (§11.4.98). UI non-introspectable parts via the §11.4.117 pixel/OCR oracle.	window-scoped MP4 + vision verdict (§11.4.159); docs/qa/<run-id>/.
Security	DNS-leak + kill-switch-fails-closed across process crash (E4/E8); session token never in logs (§11.4.10); permission-revoke teardown (E2).	leak-probe captures; log audit.
Stress	1 GB+ sustained transfer; 30-min soak; TSFN status burst (E12); rapid start/stop cycles (handle generation guard, E6).	per-iteration latency + RSS series (§11.4.85).
Chaos	kill the ability process mid-transfer (E4); revoke VPN mid-stream (E2); kill UI process, keep tunnel, relaunch + re-bind (E5); corrupt/short tun fd (E3).	categorised recovery traces; fail-closed asserted.
Performance / benchmarking	ability-process RSS vs the §10 budget (MASQUE vs plain separately); throughput vs	DevEco profiler RSS series; the §10 G-gate.

Test type	What it exercises here	Evidence / anti-bluff
Scaling	Android baseline; NAPI per-call overhead (status only). many routes / large RouteMap reconcile in the ability without RSS blowup.	RSS vs route-count curve.
UI / UX	the shared Dart screens render identically on OHOS (the UI ports for free); ConnectButton is a pure function of the relayed status stream (CI2); state announced not just colored (CI5).	visual-regression + a11y assertions [04_UI §9].
Challenges / HelixQA	a HelixQA bank entry drives the E2E journey + scores PASS only on positive captured evidence (§11.4.27).	unforgeable on-device evidence.

Paired §1.1 meta-test mutations protect each gate (e.g. strip the §9 consistency assertion → contract test must FAIL; force the relay to emit Connected while the core is Down → E2E leak/kill-switch test must FAIL). Where the SIG fork genuinely lacks an API (E9/E11), the test is an honest **SKIP-with-reason** (§11.4.3) plus a tracked migration item — never a green that masks a missing capability.

14. Open decisions surfaced by this document

ID	Decision	Options	Default (pending verification)
D-OHOS-1	cross-ability status/command IPC transport (§7)	@ohos.rpc · commonEventManager · shared KV-store	@ohos.rpc (ordering + delivery for O-I10) — UNVERIFIED
D-OHOS-2	memory-ceiling fallback rung if the ability RSS budget is breached (§10)	shrink buffers · MASQUE-off-device · split WG/QUIC	rung 1 (shrink buffers) first; escalate on G-gate failure
D-OHOS-3	OHOS Rust target triple + NDK toolchain (§12) split-tunnel field mapping	*-linux-ohos via ohos-rs / Huawei NDK	pin at Phase-3 entry — UNVERIFIED verify against DevEco SDK;

ID	Decision	Options	Default (pending verification)
D- OHOS-4	(splitExcludeApps → blockedApplications/ trustedApplications, §5/§11)	SDK-version dependent	SKIP+degrade if absent (E11)
D- OHOS-5	whether VpnExtensionAbility is guaranteed a separate process (§1)	separate / co- located	design for separate; co- located degrades gracefully — UNVERIFIED

Sources verified

- 04_VPN_CLD/HelixVPN-Architecture-Refined.md §5 (Flutter-for-8-platforms, HarmonyOS SIG fork, per-platform shims, NE memory ceiling) — accessed 2026-06-25.
- 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md §6/§6.1 (TunnelPlatform contract, "UI ports for free / shim does not", Phase-3 risk) — accessed 2026-06-25.
- final/03-client-core-and-ui.md §3 (FFI surface, frb TunnelStatus mirror + extensions), §4 (TunnelPlatform/TunnelConfig/PlatformTunnelEventKind), §5.1 (iOS C-ABI helix_core_* pattern), §5.5 (HarmonyOS row) — accessed 2026-06-25.
- final/v02-data-plane/orchestrator-and-state.md §4.1 (frozen 5-variant TunnelStatus), §4.3 (emit cadence), §4.4 (Down.reason vocab), O-I10/O-I11 — accessed 2026-06-25.
- research-flutter_ffi.md §2 (UniFFI-Dart not production-ready → NAPI/C-ABI path), §3 (cdylib packaging), §5 (OpenHarmony SIG flutter_flutter, ohos channel→HAP, Flutter OHOS baseline, ArkTS vpnExtension/VpnExtensionAbility bridged via NAPI — version specifics flagged UNVERIFIED), §6 (Riverpod StreamProvider) — accessed 2026-06-25.
- research-ios_android.md (NE memory ceiling discipline reused as the OHOS design target; VpnService in-process vs NE separate-process contrast) — accessed 2026-06-25.

UNVERIFIED markers in this document (§0, §1, §4, §5, §10, §12, §14) are the Phase-3 latest-source verification gates (§11.4.99) against the DevEco SDK + OpenHarmony Network Kit docs; none is assumed true. Per §11.4.6 they are explicitly flagged rather than guessed, and per §11.4.3 a SIG-fork capability gap resolves to an honest SKIP-with-reason, never a fabricated PASS.